

Hikaru Wakaura and Taiki Tanimae
QIRI (Quantum Integrated Research Institute Inc.)
Tokyo 107-0061, Japan
h.wakaura@deeptell.jp ORCID: 0000-0001-8381-8323

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To the Editors, *PRX Quantum*

Re: Submission of “Chaotic states outperform quantum many-body scars as codes under generic measurement: a refutation of scar-enhanced information protection”

On July 9, 2026, we submit for your consideration the manuscript above (research article, two authors) together with Supplemental Material (one PDF: eight sections, three figures, one table) and an openly available simulation package reproducing every result.

Quantum many-body scars—rare non-thermal, weakly entangled eigenstates that resist thermalization—are widely regarded as promising long-lived carriers of quantum information. We put that intuition to a quantitative test and overturn it. Encoding a logical qubit (and, more generally, the largest code a scar subspace admits) in the scar states of a monitored circuit and measuring the reference-qubit coherent information, we find that generic chaotic (thermal) states protect the encoded information *better* than scars under every measurement we examined, at every system size, in two independent models. A striking part of the story is methodological: a comparison against a single, seemingly natural thermal reference state reports the opposite conclusion, and only ensemble averaging reveals that this reference is an atypical outlier.

The message is that the resource protecting quantum information under measurement is a code’s spread in the measurement basis—scrambling—not its athermality; weak ergodicity breaking is a liability rather than an asset. The mechanism is sharp and, we believe, memorable: the emergent symmetry that makes scars special is exactly what dooms them, because its generator coherently couples the encoded states and its Casimir carries an irreducible variance that a measurement can resolve. We turn this into a falsifiable prediction and confirm it: in a second, exactly-solvable model whose scars close the algebra *exactly*, the same Casimir measurement instead protects the scar perfectly, isolating the variance as the single control parameter. Beyond the physics, the ensemble-baseline reversal is a cautionary lesson of broad utility for the many groups now benchmarking codes and channels against individual eigenstates.

We believe *PRX Quantum* is the natural home for this work. It operationally realizes, directly in the monitored-circuit setting, the correspondence between eigenstate thermalization and approximate quantum error correction; it removes weak-ergodicity-breaking subspaces from the design space of measurement-robust quantum memories; and it delivers a clean, reproducible, mechanism-level result of the conceptual significance and rigor that define the journal’s scope. All code and data are openly available for one-command reproduction.

We suggest the following referees, all leading experts in measurement-induced dynamics, quantum error correction, or quantum many-body scars (institutional contact addresses are publicly listed):

- Prof. Matthew P. A. Fisher, University of California, Santa Barbara — measurement-induced

transitions

- Prof. Ehud Altman, University of California, Berkeley — error correction and monitored circuits
- Dr. Michael J. Gullans, NIST / University of Maryland — purification transitions
- Prof. Vedika Khemani, Stanford University — random and monitored circuits
- Prof. Adam Nahum, École Normale Supérieure, Paris — entanglement dynamics
- Prof. Zlatko Papić, University of Leeds — quantum many-body scars
- Prof. Maksym Serbyn, IST Austria — quantum many-body scars
- Dr. Sanjay Moudgalya, Technical University of Munich — exact scars and spectrum-generating algebras

We do not request the exclusion of any referee.

On behalf of both authors,

Hikaru Wakaura

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